

Software Requirements Specification

Club Website and Coding Activity Platform

Compete to Compute (C2C)

Version 1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the C2C Club Website and Coding Activity Platform. It serves as the primary reference for developers, testers, stakeholders, and the club committee during system design, implementation, testing, and deployment.

It incorporates additional requirements submitted by the client, transforming the platform from a statistics-and-leaderboard site into a full member productivity and engagement platform built around a gamified dashboard, coding analytics, roadmaps, teams and a unified activity calendar.

1.2 Scope

The proposed system is a website designed to manage the club's online presence, encourage coding participation among students, and keep members engaged through gamification, analytics and structured learning paths.

The system provides:

- College email authentication
- Member profile management
- LeetCode and HackerRank account linking
- A gamified member dashboard as the post-login homepage
- XP, Levels, Badges, Achievements and Milestones
- Detailed coding analytics with heatmap and language distribution
- Coding roadmaps (DSA, CP, Web Development, AI, Android, DevOps)
- Team formation for hackathons and internal competitions
- Weekly challenges (Easy / Medium / Hard / Surprise) with automatic leaderboard updates
- A unified coding calendar for targets, events, contests, hackathons, workshops and deadlines
- Automatic point and XP calculation
- Weekly leaderboard with rank movement indicators
- Blog management
- Contest participation and rankings
- Announcements and mentor messaging
- Admin dashboard

1.3 Definitions

Term	Description
Member	Verified student using a college email
Admin	Club committee member with management privileges
Mentor	Senior member or admin who guides members and sends mentor messages
Target / Weekly Challenge	Weekly coding challenge created by Admin; may be Easy, Medium, Hard or Surprise
Leaderboard	Ranking of members based on points, with weekly rank movement
XP (Experience Points)	Points earned from solving problems, targets, contests and challenges that drive Level progression
Level	A tier derived from accumulated XP reflecting a member's overall progress
Badge	A visual reward granted automatically when a member meets a specific condition
Achievement / Milestone	A defined accomplishment (e.g. 50 Problems Solved) that unlocks a Badge
Streak	The number of consecutive days a member has logged qualifying coding activity
Roadmap	A structured, ordered set of tasks for a track such as DSA, CP, Web Development, AI, Android or DevOps
Team	A group of members created for hackathons, SIH or internal competitions
Firebase	Backend platform used for authentication, database, and hosting

1.4 References

- [Firebase Documentation](#)
- [LeetCode Public API Documentation](#)
- [HackerRank Public Endpoints](#)
- [Smart India Hackathon \(SIH\) Guidelines](#)
- [IEEE SRS Guidelines \(IEEE 830\)](#)

1.5 Document Overview

This document describes the overall system, functional requirements, external interfaces, non-functional requirements, architecture, database structure, and future enhancements.

2. Overall Description

2.1 Product Perspective

The C2C Club Website is a standalone website that integrates with Firebase, LeetCode, and HackerRank. It enables student authentication, coding activity tracking, gamified progress tracking, content management, and leaderboard generation. The post-login homepage is now a consolidated member dashboard rather than a plain statistics page.

2.2 Product Functions

The system provides the following major modules:

- Authentication
- User Profile
- Coding Account Integration
- Member Dashboard (Homepage)
- Gamification (XP, Levels, Badges, Achievements, Milestones)
- Coding Analytics
- Coding Roadmaps
- Weekly Challenges
- Points and XP Calculation
- Leaderboard
- Team Formation
- Coding Calendar
- Contest Management
- Blog Management
- Announcements and Mentor Messaging
- Admin Dashboard

2.3 User Classes

User	Permissions
Guest	Browse public pages
Member	Link coding accounts, view dashboard, analytics, roadmaps, teams, calendar and leaderboard, join challenges and teams
Mentor	All Member permissions, plus send mentor messages and hold the Mentor Badge
Admin	Manage members, blog, contests, weekly challenges, roadmaps, calendar and announcements
Super Admin	Full administrative privileges

2.4 Operating Environment

- Modern Web Browsers (Chrome, Edge, Firefox, Safari)
- Desktop, Tablet and Mobile Devices
- Firebase Authentication
- Cloud Firestore
- Firebase Cloud Functions
- Firebase Hosting

2.5 Design Constraints

- Registration is restricted to approved college email domains.
- The system relies on public LeetCode and HackerRank endpoints.
- Firebase usage is subject to service quotas.
- XP, Levels and Badges are computed server-side (Cloud Functions) to prevent client-side manipulation.
- Heatmap and analytics data are derived from synced submission data and refresh on a scheduled interval, not strictly real time.

2.6 Assumptions

- Members enter correct coding platform usernames.
- Admins manage blog, contest, roadmap, challenge and calendar content.
- Team sizes and hackathon rules are configured by Admins per event.

3. Functional Requirements

3.1 Authentication

The system shall:

- Allow signup using approved college email addresses.
- Verify email before granting access.
- Support secure login.
- Provide password reset.
- Restrict access based on user roles.

3.2 Member Profile

The system shall:

- Allow members to link LeetCode accounts.
- Allow members to link HackerRank accounts.
- Display profile statistics, current XP, Level and earned Badges.
- Allow updating or removing linked accounts.

3.3 Homepage

The Homepage serves as the public landing page of the Compete to Compute (C2C) Club Website. It provides visitors with an overview of the club and easy access to its key features.

The system shall:

- Display the club name, logo, tagline, and a brief introduction.
- Present the club's vision, mission, objectives, and achievements.
- Showcase upcoming events, workshops, contests, and recent announcements.
- Display information about the club committee, mentors, and members.
- Provide quick access to About, Events, Blog, Leaderboard, Contact, and Login pages.
- Display membership details and registration information.
- Include the club's contact information and social media links.
- Provide a responsive and user-friendly interface across all devices.

This homepage shall serve as the primary entry point to the website, enabling visitors to learn about the club and navigate to all major features.

3.4 Member Dashboard

The member dashboard replaces the plain statistics page as the homepage shown immediately after login. The system shall display, on a single consolidated screen:

- Daily coding streak, including current streak length and best streak.
- Current weekly goal and progress toward it.
- XP total and current Level, with progress toward the next Level.
- Rank movement on the leaderboard since the previous snapshot, shown as an up (↑) or down (↓) indicator with the change in position.
- Recent submissions synced from linked coding accounts.
- Upcoming contests the member is eligible for or registered in.
- Recent club announcements.
- An embedded view of the club calendar highlighting the nearest events and deadlines.
- Recent mentor messages addressed to the member.

The dashboard shall refresh automatically when new activity, announcements, or messages are received, and shall remain the default landing page for all authenticated Members, Mentors and Admins.

3.5 Gamification System

The system shall move beyond a single points value to a full gamification model:

3.5.1 XP and Levels

- Award XP for solving problems, completing weekly challenges, participating in contests, and reaching milestones.
- Maintain a configurable XP-to-Level mapping (Admin-configurable thresholds).
- Automatically promote a member's Level when the XP threshold is crossed.
- Display current Level and progress to the next Level on the profile and dashboard.

3.5.2 Badges, Achievements and Milestones

The system shall automatically detect qualifying activity and award badges, including but not limited to:

- First Accepted Problem
- 50 Problems Solved
- 100 Day Streak
- Contest Winner
- Top 10 Leaderboard
- Mentor Badge
- Allow Admins to define additional milestones and their associated badges.
- Display all earned badges on the member's profile and dashboard.
- Notify a member when a new badge or milestone is achieved.

3.6 Coding Analytics

In place of storing only linked usernames, the system shall compute and display, per member, GitHub-Insights-style analytics:

- Total problems solved.
- Breakdown of solved problems by Easy / Medium / Hard difficulty.
- Weekly progress trend.
- A submission heatmap (calendar-style, similar to a contribution graph).
- Acceptance rate.
- Contest rating (from linked platforms, where available).
- Language distribution across submitted solutions.

Analytics shall be visible on the member's own profile and, subject to privacy settings, viewable by Admins and Mentors.

3.7 Coding Roadmaps

The system shall allow members to choose a learning track and progress through it:

- Offer the following roadmap tracks: DSA, Competitive Programming (CP), Web Development, AI, Android, and DevOps.
- Allow a member to enroll in one or more roadmaps.
- Structure each roadmap as an ordered sequence of tasks that unlock progressively as prior tasks are completed.
- Track and display per-member progress through each enrolled roadmap.
- Allow Admins to create, edit and reorder roadmap tasks.
- Award XP and, where applicable, badges for roadmap milestones.

3.8 Weekly Challenges

Weekly Targets are extended into a tiered Weekly Challenges model. The system shall:

- Allow Admins to create a weekly Easy Challenge, Medium Challenge, Hard Challenge, and an optional Surprise Challenge.
- Fetch solved problems automatically from linked coding accounts to verify challenge completion.
- Calculate points and XP based on completed challenges, weighted by difficulty.
- Allow Admins to configure point and XP values per challenge tier.
- Maintain logs of calculated points and XP per member per challenge.
- Automatically update the leaderboard as challenge completions are recorded, without requiring manual recalculation.

3.9 Leaderboard

The system shall:

- Display members ranked by points and/or XP.
- Update rankings automatically as weekly challenges, contests and targets are completed.
- Maintain historical leaderboards and snapshots for rank-movement comparison.
- Compute and display each member's rank movement (↑ / ↓ / no change) relative to the previous snapshot.
- Highlight top performers.
- Hide member email addresses.

3.10 Team Formation

To support hackathons, Smart India Hackathon (SIH) and internal competitions, the system shall:

- Allow a member to create a team, specifying a name, description and optional event association.
- Allow members to browse and join open teams, subject to team size limits.
- Allow a team creator or lead to invite specific members.
- Allow invited members to accept or decline invitations.

- Allow team leads to remove members and disband the team.
- Display a member's current team memberships on their profile and dashboard.

3.11 Coding Calendar

The system shall provide a unified calendar view that consolidates:

- Weekly targets and weekly challenge windows.
- Club events.
- Coding contests.
- Hackathons.
- Workshops.
- Submission and registration deadlines.
- Allow Admins to create, edit and delete calendar entries, categorized by type.
- Allow members to filter the calendar by category.
- Surface the nearest upcoming calendar entries on the Member Dashboard.

3.12 Contests

The system shall:

- Allow admins to create and schedule coding contests.
- Allow members to register for contests.
- Track contest participation and results.
- Award points and XP based on contest performance.
- Display contest history and results.
- Grant the Contest Winner badge automatically on qualifying results.

3.13 Blog

The system shall:

- Allow admins to create, edit and delete posts.
- Support News and Journey categories.
- Display published posts.
- Save drafts.

3.14 Announcements and Mentor Messages

The system shall:

- Allow Admins to publish club-wide announcements, shown on the Member Dashboard.
- Allow Mentors to send direct messages to individual members or teams.
- Display recent announcements and mentor messages on the Member Dashboard.
- Notify members when a new announcement or mentor message is posted.

3.15 Admin Panel

The system shall:

- Manage members.
- Manage weekly challenges and legacy weekly targets.
- Manage blogs.
- Manage contests.
- Manage roadmaps and roadmap tasks.
- Manage the coding calendar (events, hackathons, workshops, deadlines).
- Manage announcements.
- Configure XP thresholds, Level tiers, badge rules and milestones.
- Assign user roles, including Mentor.
- Restrict admin access using role-based authentication.

4. External Interfaces

User Interface

Public Pages

- Home
- About
- Blog
- Leaderboard
- Contests
- Contact
- Login

Member Pages

The plain "Dashboard / Profile / Statistics" set is replaced and expanded as follows:

- Dashboard (Homepage) — streak, weekly goal, XP/Level, rank movement, recent submissions, upcoming contests, announcements, calendar snapshot, mentor messages
- Profile — linked accounts, XP, Level, Badges
- Coding Analytics — solved counts, difficulty breakdown, heatmap, acceptance rate, contest rating, language distribution
- Roadmaps — track selection and task progress
- Weekly Challenges — Easy / Medium / Hard / Surprise challenge status
- Teams — create, join and manage teams
- Coding Calendar — full calendar view
- Leaderboard

Admin Pages

- Admin Dashboard
- Member Management
- Blog Management
- Weekly Challenge Management
- Contest Management
- Roadmap Management
- Calendar Management
- Announcement Management
- Gamification Configuration (XP, Levels, Badges, Milestones)

Software Interfaces

Software	Purpose
Firebase Authentication	Login & Verification
Firestore	Database
Cloud Functions	Background processing: XP/Level computation, badge evaluation, streak tracking, heatmap and leaderboard updates
Firebase Hosting	Deployment
LeetCode	Coding activity, submissions and contest rating
HackerRank	Coding activity and submissions

Communication Interface

- HTTPS
- Email Verification
- Password Reset Emails
- In-app notifications for badges, announcements and mentor messages

5. Non-Functional Requirements

Category	Requirement
Performance	Pages load within 3 seconds; dashboard widgets render progressively
Security	HTTPS, Email Verification, Role-based Access; XP/Badge logic computed server-side to prevent tampering
Availability	99% System Availability

Category	Requirement
Scalability	Support at least 1000 users, including concurrent leaderboard and heatmap computation
Usability	Responsive UI for desktop and mobile; dashboard designed as the primary landing experience
Maintainability	Modular architecture and configurable settings, including configurable XP thresholds and badge rules
Data Consistency	Leaderboard, XP and streak data updated automatically and consistently across dashboard, profile and analytics views

6. System Architecture

Technology Stack

Layer	Technology
Frontend	React + Tailwind CSS
Backend	Firebase Cloud Functions
Database	Cloud Firestore
Authentication	Firebase Authentication
Hosting	Firebase Hosting
Scheduled Jobs	Cloud Scheduler + Cloud Functions (streak checks, leaderboard snapshots, badge evaluation)
External APIs	LeetCode & HackerRank

Architecture Flow

Users → React Frontend → Firebase Authentication → Cloud Firestore → Cloud Functions (Points / XP / Badge / Streak / Leaderboard Engine) → LeetCode / HackerRank.

7. Database Collections

Users

- uid
- name
- email
- role
- leetcodeUsername
- hackerrankUsername
- xp
- level
- currentStreak
- bestStreak
- badgels
- roadmapIds
- teamIds

Weekly Challenges

Weekly Challenges

- challengeId
- week
- tier (easy | medium | hard | surprise)
- requiredCount
- points
- xp

Points Log

- logId
- uid
- problem
- platform
- pointsAwarded
- xpAwarded

Leaderboard

- snapshotId
- week
- rankings

- previousRankings
- rankMovement

Badges

- badgeId
- name
- description
- iconUrl
- criteriaType
- criteriaValue

Member Badges

- memberBadgeId
- uid
- badgeId
- awardedAt

Roadmaps

- roadmapId
- track (DSA | CP | WebDev | AI | Android | DevOps)
- title
- tasks (ordered list)

Roadmap Progress

- progressId
- uid
- roadmapId
- completedTaskIds
- currentTaskId

Teams

- teamId
- name
- description
- eventName
- leadUid
- memberUids
- pendingInviteUids

Calendar Events

- eventId
- title
- category (target | club_event | contest | hackathon | workshop | deadline)
- startDate
- endDate
- description

Announcements

- announcementId
- title
- body
- postedBy
- postedAt

Mentor Messages

- messageId
- fromUid
- toUid / toTeamId
- body
- sentAt

Contests

- contestId
- title
- date
- description
- participants
- results

Blog Posts

- postId
- title
- category
- author
- status

Appendix

Open Items

- Exact XP values per activity type to be finalized with the club committee.
- Final list of badge criteria and icon designs to be confirmed.

Future Enhancements

- Mobile application
- Email notifications
- Blog comments
- Public team profile pages